

Lecture 06

Digital Literacy

Chapter lecture — 14 topics with explanations, worked examples, Urdu notes, practice questions and a full answer key.

BIEK Computer Science XI · Bilingual support: English + اردو

بی آئی ای کے کمپیوٹر سائنس — گیارہویں جماعت

14 book lectures in this pack

★ Golden topics, Urdu notes, review and answer key

XI

How to use this BIEK study-guide lecture

Each PDF follows one textbook lecture from the BIEK / Sindh Computer Science notes (New Sindh Curriculum). Read *Learn it*, study the SVG diagram, try the worked example on paper, then answer *Check yourself* before the answers. **Extra board practice** (MCQs, short and long questions) is added after the book content — it is not in the textbook. Career-only extras that were not in the book lecture list have been removed. Math is rendered with MathJax. ★ Golden topics are high-yield in board exams. The green box repeats the idea in Urdu.

ہر لیکچر کتاب کے درس کے مطابق ہے۔ کتاب کے بعد اضافی MCQs اور سوالات ہیں۔ جو اضافی کیریئر صفحات کتاب کی فہرست میں نہیں تھے وہ ہٹا دیے گئے ہیں۔

Lectures in this pack

Chapter 6: Digital Literacy. Study one lecture at a time, then use the chapter review and answer key at the end of this PDF.

L01. 6.1 & 6.2 Introduction to Digital Literacy

L02. 6.3 Types of Data (Qualitative vs Quantitative)

L03. 6.4 Data-Collection Strategies

L04. 6.4.1 – 6.4.2 Interviews and Surveys

L05. 6.4.3 – 6.4.5 Prototypes, Observation, and Simulations

L06. 6.5 Primary and Secondary Data

L07. 6.6 Designing a Data-Collection Approach

L08. 6.7 Presenting Data Using Digital Tools

L09. 6.7.1 – 6.7.4 Spreadsheets, Presentations, Infographics, Reports

L10. 6.8 Case Study: Digital Inquiry Project

L11. Step 1 – 3: Advanced Search & Methodology

L12. Step 4 – 6: Survey Prep & Primary Data Collection

L13. Step 7 – 9: Secondary Data & Spreadsheet Organization

L14. Step 10 – 12: Data Analysis, Conclusion & Artefact

Digital Literacy

This chapter teaches you how to find, collect, organise and present information using digital tools, safely and responsibly. You will learn about types of data, ways to collect it, and how to turn it into a useful report, presentation or infographic. At the end you will follow a complete research project from the first question to the final product.

اس باب میں آپ سیکھیں گے کہ ڈیجیٹل آلات کی مدد سے معلومات کو محفوظ اور ذمہ دارانہ طریقے سے کیسے تلاش کیا جائے، جمع کیا جائے، ترتیب دیا جائے اور پیش کیا جائے۔ آخر میں آپ ایک مکمل تحقیقی منصوبہ (Digital Inquiry Project) شروع سے آخر تک دیکھیں گے۔

What you will learn

- Define digital literacy and explain why it matters in the Information Age.
- Describe five data-collection strategies: interviews, surveys, prototypes, observation and simulations.
- Compare primary and secondary data by source, examples, purpose and control.
- Use spreadsheet formulas such as `=SUM()`, `=AVERAGE()` and `=COUNTIF()` to analyse data.
- Use advanced search operators (" ", `filetype:`, `site:`) and evaluate the reliability of a source.
- Distinguish between qualitative and quantitative data and give examples of each.
- Write clear, unbiased survey questions and recognise badly written ones.
- List the eight steps for designing a data-collection approach, including ethics and consent.
- Choose a suitable digital tool (spreadsheet, presentation, infographic, report, dashboard) to present findings.
- Carry out all 12 steps of a digital inquiry project and produce a final digital artefact.

6.1 & 6.2 Introduction to Digital Literacy

LEARN IT

Digital literacy is the ability to understand and use digital devices, the internet, and online tools in a **safe, effective, and responsible** manner.

Look at the three key words:

- **Safe** – you protect yourself and others online (for example, you do not share passwords or private details).
- **Effective** – you get the job done well (for example, you find the right information quickly).
- **Responsible** – you use information honestly and fairly (for example, you do not copy someone else's work or spread false news).

In our increasingly digital world, digital literacy is essential for **everyone**, not only for computer experts. It enables you to:

- **find** information,
- **communicate** with others,
- **produce** digital content, and
- **solve problems** through technology.

Did you know? Digital literacy is not only *using devices*; it is *using digital information safely and wisely*. A digitally literate student can:

Locate information



Organise it properly



Turn it into something useful



Share it responsibly

WORKED EXAMPLE

Is this student digitally literate?

Ali can type fast and plays games on his phone all day. When his teacher asks him to find the population of Karachi, he copies the first number he sees on a random website and does not check where it came from.

Step 1 – Can he use devices? Yes. He has basic skills.

Step 2 – Can he locate and judge information? No. He did not check whether the website was reliable.

Step 3 – Conclusion: Ali has *computer skills* but is not yet fully *digitally literate*. A digitally literate student would compare two or three trusted sources (for example, the Pakistan Bureau of Statistics website) before using the number.

REMEMBER IT

Remember the four verbs of a digitally literate student: **Locate – Organise – Create – Share** ("LOCS": think of a *lock* that keeps your digital life safe).

COMMON MISTAKE

Mistake: "Digital literacy means knowing how to use a computer." **Fix:** Using a computer is only the start. Digital literacy also means finding, judging, creating and sharing information safely and responsibly.

اردو میں سمجھیں

ڈیجیٹل خواندگی (Digital Literacy) سے مراد ڈیجیٹل آلات، انٹرنیٹ اور آن لائن ٹولز کو محفوظ، مؤثر اور ذمہ دارانہ طریقے سے سمجھنا اور استعمال کرنا ہے۔ ایک ڈیجیٹل خواندہ طالب علم معلومات تلاش کر سکتا ہے، انہیں ترتیب دے سکتا ہے، ان سے کوئی مفید چیز بنا سکتا ہے اور انہیں ذمہ داری سے دوسروں تک پہنچا سکتا ہے۔

CHECK YOURSELF

1. Write the definition of digital literacy in one sentence.
2. Name the four things digital literacy enables you to do.

Answers: 1. The ability to understand and use digital devices, the internet and online tools in a safe, effective and responsible manner. 2. Find information, communicate, produce digital content, and solve problems through technology.

LEARN IT

The **Information Age** is the period we live in now, where vast amounts of data are **produced, shared** and **accessed instantly** through the internet, mobile devices and digital platforms (websites, apps, social media).

Simply **having access** to technology is **not enough**. Owning a smartphone does not make you good at using information. You must develop the skills to **understand, manage** and **use** information wisely.

Digital literacy goes beyond basic computer skills. It involves three kinds of ability:

Ability	What it means	Example
Technical	Knowing how to operate devices and software.	Creating a Google Form; using a spreadsheet.
Cognitive (thinking)	Thinking critically about information.	Checking whether a news story is true before believing it.
Social	Communicating and behaving well with others online.	Being polite in a class WhatsApp group; respecting privacy.

Together these abilities help you function successfully in a **digital society**.

WORKED EXAMPLE

Applying the three abilities. A video claims "Drinking hot water cures dengue." What does a digitally literate student do?

Technical: Uses a search engine to search `dengue treatment site:.gov`.

Cognitive: Notices the video has no doctor or source, and that official health websites do not support the claim. Concludes it is false.

Social: Does not forward the video, and politely tells the family group that it is not reliable.

REMEMBER IT

Three abilities = "**Tech, Think, Talk**": Technical, Cognitive (think), Social (talk with others).

COMMON MISTAKE

Mistake: Thinking that "more information" automatically means "better information". **Fix:** In the Information Age the problem is not finding *some* information; it is choosing the *right, reliable* information from too much.

معلوماتی دور (Information Age) میں انٹرنیٹ اور موبائل کے ذریعے بے پناہ معلومات فوراً تیار اور شیئر ہوتی ہیں۔ صرف ٹیکنالوجی تک رسائی کافی نہیں؛ معلومات کو سمجھنے، سنبھالنے اور دانشمندی سے استعمال کرنے کی مہارت ضروری ہے۔ ڈیجیٹل خواندگی میں تکنیکی، ذہنی (Cognitive) اور سماجی صلاحیتیں شامل ہیں۔

CHECK YOURSELF

1. What is the Information Age?

2. Name the three types of ability that digital literacy involves.

Answers: 1. A period in which vast amounts of data are produced, shared and accessed instantly through the internet, mobile devices and digital platforms. 2. Technical, cognitive and social abilities.

Extra board practice

These MCQs and questions are extra to the textbook. Close the notes, attempt them, then open the key.

MCQ 1. Digital literacy is:

- a. Using digital tools safely and smartly
- b. Owning a phone only
- c. Typing 10 words
- d. Watching any video

MCQ 2. It includes:

- a. Search, evaluate, create, behave ethically
- b. Only cabling
- c. Only NAND
- d. Only sprints

MCQ 3. A digitally literate student can:

- a. Tell a reliable source from a forward
- b. Believe every screenshot
- c. Post passwords
- d. Skip citations

MCQ 4. Information age means:

- a. Value of data/knowledge is huge
- b. Stone tools
- c. No networks
- d. No computers

MCQ 5. Safety in DL includes:

- a. Passwords, privacy, respectful posts
- b. Sharing OTPs
- c. Unknown USBs always
- d. Public exam photos of others without consent

MCQ 6. This chapter later asks you to:

- a. Collect and present data yourself
- b. Build CPUs
- c. Lay fibre
- d. Write OSI sniffers

Extra short questions

1. Define digital literacy.
2. Three skills besides "knowing Word".
3. One unsafe practice.
4. Why it is in a CS course, not only Islamiyat/civics.

Extra long question

1. A cousin forwards a "BIEK result leaked" screenshot. Using digital literacy, write the steps you take before sharing, and how you would present your finding.

Extra answer key

MCQs: 1. A 2. A 3. A 4. A 5. A 6. A

Short

1. Ability to find, evaluate, use, create and share digital information safely.
2. Search operators, source judging, privacy, creating a chart, citing.
3. Using the same password for email and JazzCash.
4. You produce and analyse data — you are responsible for it.

Long

1. Check official site/date/URL, reverse-image, look for primary source. Do not share.
Present as a short report: claim, checks, conclusion. Link to later inquiry project.

6.3 Types of Data (Qualitative vs Quantitative) ★ Golden

LEARN IT

Data means facts, numbers, words or observations that we collect. The data gathered through a survey, an investigation or a classroom project can be described in two ways: **qualitatively** or **quantitatively**. Knowing the difference is critical, because each type is analysed in a different way.

6.3.1 Qualitative Data

Qualitative data describes what people **think, feel, notice or experience**. It is **non-numerical** and **descriptive**: it is made of words, not numbers. The word comes from *quality* (what something is like).

Examples: students' opinions about online classes; descriptions of study habits; observation notes ("The students looked bored after 20 minutes").

6.3.2 Quantitative Data

Quantitative data deals with **numbers**. The word comes from *quantity* (how much or how many). It is useful when you need to **count, measure, compare, calculate averages**, or **display results in charts**.

Examples: number of students using the internet daily; marks obtained in a test; time spent online in hours.

6.3.3 Comparing Qualitative and Quantitative Data

Type of data	Nature	Examples	Best used for
Qualitative	Descriptive, non-numerical	Opinions, reasons, behaviour	Understanding why and how something happens
Quantitative	Numerical, countable or measurable	Hours, scores, percentages, totals	Comparing results and making graphs/charts

LEARN IT

Going a little further (background). Quantitative data is sometimes split into two kinds:

- **Discrete** data can only take separate, countable values, e.g. number of students in a class (35, 36 – never 35.5).
- **Continuous** data can take any value in a range and is *measured*, e.g. height (162.4 cm) or time online (2.75 hours).

Also note: a question with fixed categories (e.g. "Which activity do you use your phone for most? Lectures / Notes / Social media / Games") gives **categories** that you can **count**. In this chapter's case study such answers are treated as **quantitative (categories)**, because you count how many students chose each option.

WORKED EXAMPLE

Classify each piece of data.

Data	Type	Reason
"I like online classes because I can replay lectures."	Qualitative	An opinion with a reason, in words.
Ayesha scored 78 out of 100.	Quantitative	A score – a number you can compare.
28 out of 40 students use the internet daily.	Quantitative	A count (and gives 70%).
Observer notes: "Students chat during the last period."	Qualitative	A description of behaviour.
Bilal spends 3.5 hours online per day.	Quantitative	A measurement of time.

Quick test: ask "Can I calculate an average of this?" If yes, it is quantitative. If the answer is words, it is qualitative.

REMEMBER IT

QUALitative = **QUAL**ity = words and feelings. **QUANT**itative = **QUANT**ity = numbers and counts. Qualitative answers **why/how**; quantitative answers **how many/how much**.

COMMON MISTAKE

Mistake: Thinking that any data containing a number is quantitative. A phone number or a roll number is a *label*, not a quantity – averaging roll numbers is meaningless. **Fix:** Data is quantitative only if the number measures or counts something.

کیفیتی ڈیٹا (Qualitative Data) الفاظ پر مشتمل ہوتا ہے اور بتاتا ہے کہ لوگ کیا سوچتے، محسوس کرتے یا تجربہ کرتے ہیں، جیسے آن لائن کلاسز کے بارے میں رائے۔ مقداری ڈیٹا (Quantitative Data) اعداد پر مشتمل ہوتا ہے، جیسے ٹیسٹ کے نمبر یا آن لائن گزارے گئے گھنٹے۔ کیفیتی ڈیٹا "کیوں اور کیسے" سمجھنے کے لیے، جبکہ مقداری ڈیٹا موازنہ کرنے اور چارٹ بنانے کے لیے بہترین ہے۔

EXAM TIP

Common questions: "Differentiate between qualitative and quantitative data" (4–5 marks) or "Classify the following data". Write the answer as a **table** with four rows: **nature**, **examples**, **best used for**, plus a one-line definition of each. Always give **two examples** of each type. The source's table (Nature / Examples / Best used for) is exactly what examiners expect.

CHECK YOURSELF

1. Is "time spent online in hours" qualitative or quantitative?
2. Which type of data is best for understanding *why* something happens?
3. Classify: "The canteen food is too oily."

Answers: 1. Quantitative (measurable number). 2. Qualitative. 3. Qualitative (an opinion in words).

Extra board practice

These MCQs and questions are extra to the textbook. Close the notes, attempt them, then open the key.

MCQ 1. Qualitative data is:

- a. Categories / words (favourite subject)
- b. Heights in cm only
- c. Always continuous numbers
- d. Always bits

MCQ 2. Quantitative data is:

- a. Numeric measurements or counts
- b. Colours of bags only
- c. Always opinions
- d. Always URLs

MCQ 3. Favourite colour is:

- a. Qualitative
- b. Quantitative continuous
- c. A primary key always
- d. A gate

MCQ 4. Height is:

- a. Quantitative continuous
- b. Qualitative
- c. Boolean only
- d. Tertiary

MCQ 5. Number of siblings is:

- a. Quantitative discrete
- b. Qualitative
- c. Analog voltage
- d. A query

MCQ 6. Why the type matters:

- a. It decides the chart and statistic
- b. It decides OSI layer
- c. It decides NAND grouping
- d. It decides Agile sprint length

Extra short questions

1. Define qualitative vs quantitative with examples.
2. Discrete vs continuous quantitative.
3. Which chart for qualitative shares?
4. Which average for qualitative?

Extra long question

1. A survey has: gender, hours of study, favourite app, test marks. Classify each, pick a chart, and say mean/median/mode which is legal.

Extra answer key

MCQs: 1. A 2. A 3. A 4. A 5. A 6. A

Short

1. Qual: labels (blood group). Quant: numbers (marks).

2. Discrete: counts (students). Continuous: measurements (time, temperature).
3. Bar or pie (few categories).
4. Mode — mean of “red/blue” is meaningless.

Long

1. Gender qual → pie/bar, mode. Hours quant cont → hist, mean/median. App qual → bar, mode. Marks quant → hist/box, mean/median. Table.

6.4 Data-Collection Strategies ★ Golden

LEARN IT

Digital literacy is not limited to finding information that **already exists** on the internet. You should also learn how to gather **original** (new) information yourself. A **data-collection strategy** is a planned method for gathering this information.

The five common strategies are:

Interviews

Surveys

Prototypes

Observation

Simulations

Strategy	How it works (one line)	Usual data type	Best when...
Interview	Planned conversation with a person or small group	Mostly qualitative	You need deep, detailed answers from a few people
Survey	Same questions asked to many people	Quantitative and qualitative	You need answers from many people quickly
Prototype	Users test an early version and give feedback	Both	You want to improve a product or solution
Observation	Watch and record; no questions asked	Notes (qualitative) or counts (quantitative)	You want to see what people actually do
Simulation	Computer model; change variables and see what happens	Mostly quantitative	Real experiments are costly, risky or difficult

REMEMBER IT

Mnemonic for the five strategies: **"I S P O S"** – **"I Saw People On Screens"**: Interviews, Surveys, Prototypes, Observation, Simulations.

اردو میں سمجھیں

ڈیجیٹل خواندگی صرف موجود معلومات تلاش کرنے تک محدود نہیں، بلکہ نئی معلومات خود جمع کرنا بھی اس کا حصہ ہے۔ ڈیٹا جمع کرنے کے پانچ عام طریقے یہ ہیں: انٹرویو، سروے، پروٹوٹائپ، مشاہدہ (Observation) اور سمیولیشن (Simulation)۔

EXAM TIP

"List and explain the data-collection strategies" is a frequent long question. Name all **five**, give a one-sentence definition and **one example** for each, and say what type of data each gives. Detailed notes on each follow in the next two topics.

CHECK YOURSELF

1. Name the five data-collection strategies.
2. What does "original information" mean?

Answers: 1. Interviews, surveys, prototypes, observation, simulations. 2. New information that you gather yourself, rather than information that already exists.

Extra board practice

These MCQs and questions are extra to the textbook. Close the notes, attempt them, then open the key.

MCQ 1. Data-collection strategies include:

- a. Interview, survey, prototype, observation, simulation
- b. Only NAND
- c. Only OSI ping
- d. Only K-maps

MCQ 2. You collect data when:

- a. Existing sources are not enough
- b. You already have the full census always
- c. The question is undefined
- d. You only need a meme

MCQ 3. A strategy must match:

- a. The question and the people
- b. The teacher's favourite chart only
- c. Agile always
- d. TCP window

MCQ 4. Bias in collection:

- a. Systematically distorts answers
- b. Improves SD always
- c. Is required in exams
- d. Is a primary key

MCQ 5. Ethics: people should:

- a. Know and consent
- b. Be filmed secretly always
- c. Hand over passwords
- d. Be named in viral posts

MCQ 6. This is ★ because papers ask:

- a. Name + when to use + one limit of a method
- b. Only Python turtles
- c. Only ER
- d. Only Big O

Extra short questions

1. List five strategies from the book.
2. One advantage of planning methods first.
3. One ethical rule.
4. What is a biased question?

Extra long question

1. Choose methods for: (a) why students skip practicals (b) how a new ID-card prototype feels (c) traffic at the gate. Justify. ★ Golden.

Extra answer key

MCQs: 1. A 2. A 3. A 4. A 5. A 6. A

Short

1. Interview, survey, prototype, observation, simulation (and others in 6.4).
2. You do not collect the wrong type of data for your chart later.
3. Informed consent; anonymise students.
4. "Don't you agree our canteen is terrible?" — pushes the answer.

Long

1. (a) interview+survey (b) prototype test (c) observation/count or simulation. Limits of each. Ethics sentence.

6.4.1 – 6.4.2 Interviews and Surveys ★ Golden

6.4.1 Interviews

LEARN IT

An **interview** is a **planned conversation** used to gather information **directly from a person**.

- It may be **one-to-one** (you and one person) or in a **small group**.
- Questions are **planned in advance**, so you do not forget anything important.
- Answers are usually **qualitative** (opinions, reasons, experiences).

Strength: you get deep, detailed answers and can ask follow-up questions ("Why do you say that?"). **Weakness:** it takes a lot of time, so you can only interview a few people.

6.4.2 Surveys / Questionnaires

LEARN IT

A **survey** (or **questionnaire**) is useful when the **same set of questions** needs to be asked to **many people**.

- It may be on **paper** or created with **digital tools** such as **Google Forms** or **Microsoft Forms**.
- **Note:** online survey tools **automatically collect responses** and **display basic charts**, which makes data analysis much easier.

Types of survey and interview questions

Question type	What it looks like	Data it gives
Closed: Yes/No	Do you study with your phone nearby? (Yes / No)	Countable (quantitative)
Closed: Multiple choice	Which activity do you use your phone for most? (Lectures / Notes / Social media / Games)	Categories you can count
Number / scale	How many hours do you use your phone daily? ____	Quantitative
Rating scale	"My phone helps me study." 1 = Strongly disagree ... 5 = Strongly agree	Quantitative (you can average the ratings)
Open-ended	How does your phone help you in studies?	Qualitative (free text)

Good questions vs bad questions

A good question is **clear, short, unbiased** (it does not push the person towards one answer) and asks about **one thing at a time**.

Problem	Bad question	Better question
Leading / biased – suggests the "right" answer	Don't you agree that phones waste students' time?	How do phones affect your study time? (Helps a lot / Helps a little / No effect / Harms a little / Harms a lot)
Double-barrelled – two questions in one	Do you use your phone for lectures and games?	Split into two: "Do you use your phone for lectures?" and "Do you use your phone for games?"
Vague – unclear words	Do you use your phone a lot?	How many hours do you use your phone on a normal school day?
Overlapping options	1–3 hrs / 3–5 hrs / 5+ hrs (where does exactly 3 go?)	Less than 3 hrs / 3 to less than 5 hrs / 5 hrs or more
Sensitive / too personal	What is your father's monthly income?	Leave it out unless it is truly needed for the study.

WORKED EXAMPLE

Building a short Google Form, step by step. Research question: "How do Class XI students travel to college?"

Step 1: Open Google Forms → Blank form → Title: "Travel to College Survey". Add a description: "This survey is anonymous and for a class project. Do you agree to take part? (Yes/No)" – this is your **consent** question.

Step 2: Q1 (multiple choice): "How do you usually travel to college?" Walk / Bus / Rickshaw / Motorbike / Car / Other.

Step 3: Q2 (number): "How many minutes does your journey take?"

Step 4: Q3 (open-ended): "What is the biggest problem you face while travelling?"

Step 5: Click *Send* and share the link in the class group. Responses appear in the *Responses* tab with automatic pie/bar charts, and can be opened in Google Sheets.

Result: Q1 and Q2 give quantitative data; Q3 gives qualitative data.

REMEMBER IT

Interview = In-depth with few. Survey = Same questions to many. Think of a doctor talking to one patient (interview) vs a census form sent to every house (survey).

COMMON MISTAKE

Mistake: Writing only open-ended questions in a survey for 100 people. You will get 100 paragraphs that are very hard to analyse. **Fix:** Use mostly closed or number questions in a large survey, and only one or two open-ended questions.

اردو میں سمجھیں

انٹرویو ایک منصوبہ بند گفتگو ہے جس میں کسی ایک فرد یا چھوٹے گروہ سے براہ راست معلومات حاصل کی جاتی ہیں؛ اس کے جوابات عموماً کیفیتی ہوتے ہیں۔ سروے میں ایک ہی سوالات بہت سے لوگوں سے پوچھے جاتے ہیں، کاغذ پر یا گوگل فارمز (Google Forms) جیسے ڈیجیٹل ٹولز کے ذریعے، جو جوابات خودکار طور پر جمع کر کے چارٹ بھی دکھاتے ہیں۔ اچھا سوال واضح، مختصر، غیر جانبدار اور ایک وقت میں ایک ہی بات کے بارے میں ہوتا ہے۔

EXAM TIP

You may be asked to "differentiate between an interview and a survey" or to "identify what is wrong with this survey question and rewrite it". For rewriting, name the problem (leading, double-barrelled, vague, overlapping) and give the corrected version.

CHECK YOURSELF

1. Name two digital tools for creating surveys.
2. What is wrong with: "Do you enjoy maths and physics?"
3. Are interview answers usually qualitative or quantitative?

Answers: 1. Google Forms and Microsoft Forms. 2. It is double-barrelled (two subjects in one question); ask about each subject separately. 3. Usually qualitative.

Extra board practice

These MCQs and questions are extra to the textbook. Close the notes, attempt them, then open the key.

MCQ 1. An interview is:

- a. A spoken Q&A, often fewer people, deeper
- b. A 500-person tick sheet always
- c. A NAND
- d. A pie

MCQ 2. A survey is:

- a. The same questions to many people
- b. Always one celebrity chat
- c. A sort
- d. A cable

MCQ 3. Closed questions are easier to:

- a. Count and chart
- b. Explore feelings
- c. Build CPUs
- d. Route IP

MCQ 4. Open questions are better for:

- a. Reasons and quotes
- b. Instant averages
- c. Primary keys
- d. Gray codes

MCQ 5. A leading question is:

- a. Biased
- b. Required
- c. A foreign key
- d. A sprint

MCQ 6. Sampling matters because:

- a. Only asking prefects is not the whole college
- b. n always equals 1
- c. Charts ignore n
- d. Ethics forbid samples

Extra short questions

1. Differentiate interview and survey.
2. Two question-writing rules.
3. When mix both?
4. What is a pilot?

Extra long question

1. Write 4 survey questions (2 closed, 2 open) on screen time, plus 3 interview prompts. Mark any bias you avoided.

Extra answer key

MCQs: 1. A 2. A 3. A 4. A 5. A 6. A

Short

1. Interview: deep, slow, few. Survey: standardised, many, shallower.
2. One idea per item; avoid leading; simple Urdu/English; pilot test.
3. Survey for numbers, then interview to explain a surprising number.
4. Try the form on 5 classmates and fix confusing items.

Long

1. Closed: hours bands; devices. Open: why night use; effect on sleep. Interview: walk me through yesterday evening. No "don't you think phones are evil".

6.4.3 – 6.4.5 Prototypes, Observation, and Simulations ★ Golden

6.4.3 Prototypes

LEARN IT

A **prototype** is a **simple early version** of a product, model or solution. It allows an idea to be **tested before** time and effort are spent on the final version. Users **try** the prototype and give **feedback**, and this feedback becomes useful **data for improvement**.

Example: Before building a full school app, you draw the screens on paper (or make a few clickable screens in a design tool) and ask five classmates to try "finding the timetable". You note where they get confused.

6.4.4 Observation

LEARN IT

In **observation**, the researcher **watches** what people do or what happens in a situation and **records the results carefully**. **No direct questions are asked**.

The data may be **qualitative** (written notes, e.g. "students push in the queue") or **quantitative** (counts, e.g. "42 students bought samosas in 10 minutes").

Why use it? People sometimes say one thing and do another. Observation shows what really happens.

6.4.5 Simulations

LEARN IT

A **simulation** is a **computer-based model of a real situation**. **Variables** (the things that can change, like speed, temperature or number of people) can be changed to see what happens.

Simulations are highly useful when **real experiments are costly, risky or difficult** to perform.

Examples: a flight simulator for training pilots (real crashes are dangerous); a traffic simulation to test what happens if a new signal is added (changing real roads is costly); a weather model to predict a cyclone (you cannot experiment with real weather).

WORKED EXAMPLE

Choosing the right method. Which method would you use?

(a) "How many cars cross an intersection at 8 AM?" → Observation. Stand at the corner from 8:00 to 8:15 and tally each car. Asking drivers would be impossible and unnecessary; you just count.

(b) "How do students feel about the new cafeteria food?" → Survey or interview. Feelings are inside people's heads, so you must ask them. Use a survey for many students, or interviews for deeper reasons.

(c) "Will a new college website menu be easy to use?" → Prototype. Build a rough version and let users try it.

(d) "What happens to traffic if the bridge is closed for repairs?" → Simulation. Closing a real bridge to test is costly and disruptive.

REMEMBER IT

Watch it → Observation. **Ask** it → Survey/Interview. **Try** it → Prototype. **Model** it → Simulation.

COMMON MISTAKE

Mistake: Saying that observation involves asking people questions. **Fix:** In observation, **no direct questions** are asked – the researcher only watches and records. Also, a prototype is *not* the finished product; it is an early test version.

اردو میں سمجھیں

پروٹوٹائپ کسی پروڈکٹ یا حل کا ابتدائی سادہ نمونہ ہے جسے صارفین آزما کر رائے دیتے ہیں تاکہ اسے بہتر بنایا جاسکے۔ مشاہدے (Observation) میں محقق کوئی سوال نہیں پوچھتا بلکہ صرف دیکھ کر نتائج درج کرتا ہے۔ سمیولیشن کسی حقیقی صورتحال کا کمپیوٹر ماڈل ہے، جو اس وقت مفید ہے جب حقیقی تجربہ مہنگا، خطرناک یا مشکل ہو۔

EXAM TIP

Scenario questions are popular: "Which data-collection method is most suitable for ... ? Give a reason." Always name the method **and** justify it using its key feature (e.g. "Simulation, because a real experiment would be risky").

CHECK YOURSELF

1. Why would an engineer use a simulation instead of a real test?
2. Can observation produce quantitative data? Give an example.
3. What becomes the "data" when using a prototype?

Answers: 1. Because the real experiment may be costly, risky or difficult. 2. Yes – for example, counting the number of cars passing a point. 3. The feedback users give after trying the prototype.

Extra board practice

These MCQs and questions are extra to the textbook. Close the notes, attempt them, then open the key.

MCQ 1. A prototype here is:

- a. An early model to test an idea
- b. The final app store release
- c. A primary key
- d. A histogram

MCQ 2. Observation means:

- a. Watching what people actually do
- b. Only asking them
- c. Only simulating
- d. Only Googling

MCQ 3. Simulation is:

- a. A model of a system you can run
- b. A real flood in the campus
- c. A NAND chip
- d. A foreign key

MCQ 4. People say they wait 2 minutes but observation shows 10: trust:

- a. Observation of that behaviour
- b. The boast
- c. Neither ever
- d. OSI

MCQ 5. Paper screens of an app are a:

- a. Low-fidelity prototype
- b. MVP already selling
- c. Database schema
- d. Pie chart

MCQ 6. Simulations are useful when:

- a. Real experiments are costly or unsafe
- b. You already have all answers
- c. $n=1$ always
- d. Ethics forbid models

Extra short questions

1. Define prototype, observation, simulation in one line each.
2. One limit of observation.
3. One limit of simulation.
4. When is a prototype better than a survey?

Extra long question

1. A team wants a quieter library. Combine prototype, observation and simulation in a plan with ethics.

Extra answer key

MCQs: 1. A 2. A 3. A 4. A 5. A 6. A

Short

1. Prototype: early model. Observation: watch. Simulation: run a model.
2. Hawthorne effect — people act differently when watched.
3. Wrong assumptions → pretty but false results.
4. When you need to see if a design is usable, not just popular in words.

Long

1. Observe noise sources; prototype signage/zones; simulate seating. Consent to watch; no filming faces for TikTok. What would change your mind (data).

6.5 Primary and Secondary Data ★ Golden

LEARN IT

Whenever you use data, it is important to know **where it came from**. The source affects how much **control** the researcher had over collecting it, and how **confidently** the results can be used.

6.5.1 Primary Data

Primary data is collected **first-hand** by the researcher for a **particular purpose**. It is gathered **directly** from people, events or situations, rather than taken from an already published source.

Sources: interviews, surveys, observations, experiments.

6.5.2 Secondary Data

Secondary data has **already been collected by someone else** and is being used again for a **new purpose**. It is taken from existing archives (stored collections of information).

Sources: books, research articles, official reports, web statistics, online databases.

Primary Data

Fresh, direct collection
(You do the work)

Surveys · Interviews · Observation

Secondary Data

Existing archives
(Someone else did it)

Books · Reports · Databases

Primary data is collected by you; secondary data was collected by others.

6.5.3 Difference Between Primary and Secondary Data

Aspect	Primary Data	Secondary Data
Source	Collected directly by the researcher.	Already collected by others.
Examples	Survey responses, interviews, observations.	Reports, articles, websites, databases.
Purpose	Prepared for a specific study.	Reused for a new purpose.
Control	High control over questions and collection process.	Limited control because the data already exists.

WORKED EXAMPLE

Primary or secondary? Hira is researching "Internet use among students in Hyderabad".

(a) She sends a Google Form to 50 students in her college. → **Primary** (she collected it herself, for this study).

(b) She uses a PTA (Pakistan Telecommunication Authority) report on the number of broadband users in Pakistan. → **Secondary** (PTA collected it for its own purpose; Hira reuses it).

(c) She counts how many students use phones in the library during one hour. → **Primary** (her own observation).

(d) She quotes a statistic from a research article. → **Secondary**.

Why use both? Her primary data tells her about *her own* college; the secondary data lets her **compare** it with the national picture.

REMEMBER IT

Primary = "1st hand" (you collect it first). **Secondary** = "2nd hand" (like a second-hand book – someone else used it first).

COMMON MISTAKE

Mistake: Thinking data from a website is always primary because "I found it myself". **Fix:** What matters is *who collected* the data. If someone else collected it and you only found it, it is secondary.

اردو میں سمجھیں

بنیادی ڈیٹا (Primary Data) محقق خود کسی خاص مقصد کے لیے براہ راست جمع کرتا ہے، مثلاً سروے، انٹرویو یا مشاہدہ۔ ثانوی ڈیٹا (Secondary Data) پہلے ہی کسی اور نے جمع کیا ہوتا ہے اور اسے نئے مقصد کے لیے دوبارہ استعمال کیا جاتا ہے، جیسے کتابیں، رپورٹس اور آن لائن ڈیٹا بیس۔ بنیادی ڈیٹا پر محقق کا کنٹرول زیادہ اور ثانوی ڈیٹا پر کم ہوتا ہے۔

EXAM TIP

"Differentiate between primary and secondary data" is one of the most-asked questions in this chapter. Draw the four-row table: **Source, Examples, Purpose, Control**. Learn it exactly. Add one example from each side to secure full marks.

CHECK YOURSELF

1. Which type of data gives the researcher high control?
2. Is a government census report primary or secondary data for your project?
3. Give two sources of primary data.

Answers: 1. Primary data. 2. Secondary (someone else collected it). 3. Any two: interviews, surveys, observations, experiments.

Extra board practice

These MCQs and questions are extra to the textbook. Close the notes, attempt them, then open the key.

MCQ 1. Primary data is:

- a. Collected by you for this question
- b. Copied from a textbook table always
- c. Always tertiary
- d. A NAND output

MCQ 2. Secondary data is:

- a. Collected by others, reused
- b. Your interview today
- c. Always fake
- d. A pie only

MCQ 3. You control quality more with:

- a. Primary
- b. Secondary always
- c. Tertiary always
- d. None

MCQ 4. Secondary is often:

- a. Faster and cheaper
- b. Always more accurate
- c. Always ethical-free
- d. Always qualitative

MCQ 5. Census tables you download are:

- a. Secondary for you
- b. Primary for you
- c. Not data
- d. A query object

MCQ 6. A common exam pair is:

- a. Define both + one difference + one example each
- b. Only Big O
- c. Only ER
- d. Only Agile

Extra short questions

1. Table: primary vs secondary on source, control, cost, example.
2. One risk of secondary.
3. One risk of primary.
4. Can a project use both?

Extra long question

1. For "canteen waiting time", list primary and secondary data you would gather, and how each could lie. ★ Golden.

Extra answer key

MCQs: 1. A 2. A 3. A 4. A 5. A 6. A

Short

1. You vs others; high vs low control; high vs low cost; survey vs PBS stats.
2. Out of date, different definitions, hidden bias.
3. Small sample, poor questions, time.
4. Yes — and the book's inquiry project does.

Long

1. Primary: observe queues, survey. Secondary: last year's complaint log, news on food inflation. Lies: observed only at 8am; old log incomplete. Triangulate.

6.6 Designing a Data-Collection Approach

LEARN IT

Before collecting data, a researcher must have a **plan**. A plan makes sure that the information gathered is **accurate**, **relevant** (related to the question) and **useful**. Follow these eight steps in order:

1. **Define the research question clearly.** What exactly do you want to find out?
2. **Decide the type of data needed:** quantitative, qualitative, or both.
3. **Select suitable methods** (survey, interview, observation).
4. **Choose the sample or target group** – the people who will give you data.
5. **Prepare the tools** such as forms, questions and observation sheets.
6. **Collect the data** carefully and accurately.
7. **Store the data digitally** in a spreadsheet and **keep a backup**.
8. **Follow ethical practice and respect privacy. Consent is mandatory** (you must get permission).



WORKED EXAMPLE

Applying the eight steps. Topic: water wastage in your college.

1. **Question:** "How much water is wasted at the college taps during break time, and why?"
2. **Data type:** Both – quantitative (number of taps left running, minutes) and qualitative (reasons students give).
3. **Methods:** Observation (count running taps) + a short survey (ask about habits).
4. **Sample:** All 12 taps in the college; 30 students from different classes for the survey.
5. **Tools:** An observation sheet with columns *Tap no.* / *Time* / *Left running?* (Y/N) / *Minutes*; a Google Form with 5 questions.
6. **Collect:** Observe during break for five days; send the form link.
7. **Store:** Enter the observation counts into Google Sheets; download a copy to a USB drive as backup.
8. **Ethics:** Ask the principal's permission; tell students the survey is anonymous and voluntary; do not record names.

REMEMBER IT

Remember the steps with: "Quiet Teachers Make Students Think, Check, Save Ethically"
→ Question, Type, Method, Sample, Tools, Collect, Store, Ethics.

COMMON MISTAKE

Mistake: Starting to collect data before writing a clear research question. You then collect lots of data that does not answer anything. **Fix:** Always write the question first. Another mistake: keeping data in only one place – if the file is lost, the project is lost. Always keep a backup.

اردو میں سمجھیں

ڈیٹا جمع کرنے سے پہلے منصوبہ بندی ضروری ہے: تحقیقی سوال واضح کریں، ڈیٹا کی قسم طے کریں، مناسب طریقہ چنیں، نمونہ (Sample) منتخب کریں، فارم اور سوالات تیار کریں، احتیاط سے ڈیٹا جمع کریں، اسے اسپریڈشیٹ میں محفوظ کر کے بیک اپ رکھیں، اور اخلاقیات کا خیال رکھیں۔ اجازت (Consent) لینا لازمی ہے۔

CHECK YOURSELF

1. What is the first step in designing a data-collection approach?
2. Which step says "consent is mandatory"?
3. Where should collected data be stored?

Answers: 1. Define the research question clearly. 2. Step 8 – follow ethical practice and respect privacy. 3. Digitally, in a spreadsheet, with a backup copy.

Extra board practice

These MCQs and questions are extra to the textbook. Close the notes, attempt them, then open the key.

MCQ 1. Data-collection strategies include:

- a. Interview, survey, prototype, observation, simulation
- b. Only NAND
- c. Only OSI ping
- d. Only K-maps

MCQ 2. You collect data when:

- a. Existing sources are not enough
- b. You already have the full census always
- c. The question is undefined
- d. You only need a meme

MCQ 3. A strategy must match:

- a. The question and the people
- b. The teacher's favourite chart only
- c. Agile always
- d. TCP window

MCQ 4. Bias in collection:

- a. Systematically distorts answers
- b. Improves SD always
- c. Is required in exams
- d. Is a primary key

MCQ 5. Ethics: people should:

- a. Know and consent
- b. Be filmed secretly always
- c. Hand over passwords
- d. Be named in viral posts

MCQ 6. This is ★ because papers ask:

- a. Name + when to use + one limit of a method
- b. Only Python turtles
- c. Only ER
- d. Only Big O

Extra short questions

1. List five strategies from the book.
2. One advantage of planning methods first.
3. One ethical rule.
4. What is a biased question?

Extra long question

1. Choose methods for: (a) why students skip practicals (b) how a new ID-card prototype feels (c) traffic at the gate. Justify. ★ Golden.

Extra answer key

MCQs: 1. A 2. A 3. A 4. A 5. A 6. A

Short

1. Interview, survey, prototype, observation, simulation (and others in 6.4).
2. You do not collect the wrong type of data for your chart later.
3. Informed consent; anonymise students.
4. "Don't you agree our canteen is terrible?" — pushes the answer.

Long

1. (a) interview+survey (b) prototype test (c) observation/count or simulation. Limits of each. Ethics sentence.

6.7 Presenting Data Using Digital Tools

LEARN IT

Collecting data is only **one part** of the work. You must also **present** your findings so that others can understand them **quickly and easily**. Digital tools make this much easier. The four main tools are spreadsheets, presentations, infographics and reports.

6.7.1 Spreadsheets

LEARN IT

Spreadsheets such as **MS Excel**, **Google Sheets** and **Apple Numbers** arrange information in **rows** (horizontal, numbered 1, 2, 3...) and **columns** (vertical, lettered A, B, C...). The box where a row and column meet is a **cell**, named by its column and row, e.g. **B2**. With a spreadsheet you can:

- **Enter data** systematically (one record per row).
- **Calculate** totals, averages and percentages.
- **Sort** (arrange in order) and **filter** (show only some rows) records.
- **Create charts and graphs** automatically.

A **formula** always starts with =. A **range** such as B2:B4 means "cells B2 to B4".

Formula	What it does
=SUM(B2:B4)	Adds all values in B2 to B4.
=AVERAGE(B2:B4)	Mean: sum ÷ number of values.
=MAX(B2:B4) / =MIN(B2:B4)	Largest / smallest value.
=MEDIAN(B2:B4)	Middle value when sorted.
=COUNT(B2:B4)	Counts cells containing numbers.
=COUNTA(C2:C4)	Counts cells that are not empty (text or numbers).
=COUNTIF(C2:C4, "Yes")	Counts cells that match a condition.

WORKED EXAMPLE

Analysing the case-study spreadsheet. The data from the 6.8 case study (Step 8) is entered like this:

	A	B	C	D	E
1	Student	Hours	Study Use	Main Activity	Distracted?
2	1	3	Yes	Lectures	Sometimes
3	2	5	No	Social Media	Yes
4	3	2	Yes	Notes	No

Step 1 – Total hours: =SUM(B2:B4) → 3 + 5 + 2 = **10**.

Step 2 – Average hours: =AVERAGE(B2:B4) → 10 ÷ 3 = **3.33** hours (to 2 decimal places).

Step 3 – Most and least: =MAX(B2:B4) → **5**; =MIN(B2:B4) → **2**.

Step 4 – Median: sorted values 2, 3, 5 → middle = **3**. =MEDIAN(B2:B4) gives 3.

Step 5 – How many use the phone for study? =COUNTIF(C2:C4, "Yes") → **2**.

Step 6 – Percentage: =COUNTIF(C2:C4, "Yes")/COUNTA(C2:C4) → 2 ÷ 3 = 0.667; format the cell as % → **66.7%**.

Step 7 – How many said "Yes" to distracted? =COUNTIF(E2:E4, "Yes") → **1** (only "Yes" matches; "Sometimes" does not).

Step 8 – Sort: select the data → Data → Sort by column B (largest to smallest) → order becomes Student 2 (5), Student 1 (3), Student 3 (2).

6.7.2 Presentations

LEARN IT

Presentation software such as **PowerPoint**, **Google Slides** and **Canva** helps you explain findings to an **audience** in an organised way. Good practice:

- Begin with a clear **title** and the **research question**.
- Use **short bullet points** instead of long paragraphs.
- Include **tables, charts and key findings**.
- Keep the design **simple and readable** (large font, few colours).

Spreadsheets and presentations work together: you make a chart from the spreadsheet data and insert it into the slides.

Spreadsheet data



Chart / Graph



Insert



Slides

REMEMBER IT

Presentation rule of thumb: **one idea per slide**, and no more than about **six short bullet points**. If you are reading long sentences off the slide, the slide has too much text.

COMMON MISTAKE

Mistake: Typing a result by hand (e.g. writing "3.33" in a cell) instead of using a formula. If the data changes, the typed result becomes wrong. **Fix:** Always use formulas like `=AVERAGE(B2:B4)` so results update automatically. Also remember every formula must begin with `=`.

اردو میں سمجھیں

ڈیٹا جمع کرنے کے بعد اسے پیش کرنا بھی ضروری ہے۔ اسپریڈشیٹ (جیسے ایکسل اور گوگل شیٹس) میں ڈیٹا قطاروں اور کالموں میں ترتیب دیا جاتا ہے، اور `=SUM()` اور `=AVERAGE()` جیسے فارمولوں سے مجموعہ، اوسط اور فیصد نکالے جاتے ہیں۔ پریزنٹیشن سافٹ ویئر (جیسے پاور پوائنٹ) میں مختصر نکات، چارٹس اور سادہ ڈیزائن کے ساتھ نتائج سامعین کے سامنے پیش کیے جاتے ہیں۔

CHECK YOURSELF

1. Write the formula to find the average of cells D2 to D20.
2. Name three spreadsheet programs.
3. Give two rules for a good presentation.

Answers: 1. `=AVERAGE(D2:D20)` . 2. MS Excel, Google Sheets, Apple Numbers. 3. Any two: start with a clear title and research question; use short bullet points; include tables/charts/key findings; keep design simple and readable.

Extra board practice

These MCQs and questions are extra to the textbook. Close the notes, attempt them, then open the key.

MCQ 1. Spreadsheets are good for:

- a. Tables, calculations, simple charts
- b. 3D games
- c. NAND simulation
- d. OSI sniffing

MCQ 2. A presentation (slides) is for:

- a. Talking an audience through findings
- b. Storing the only copy of raw data
- c. Primary keys
- d. Kernel drivers

MCQ 3. An infographic is:

- a. A visual one-pager of facts
- b. A 40-page appendix
- c. A binary search
- d. A foreign key

MCQ 4. A report is:

- a. Structured writing with method, results, limits
- b. Only memes
- c. Only code
- d. Only ER

MCQ 5. Choose the tool by:

- a. Audience and purpose
- b. Whatever is newest
- c. Always infographic
- d. Always 80 slides

MCQ 6. Raw 2000-row sheets in a 5-minute talk are:

- a. A bad present
- b. Ideal
- c. Required by BIEK
- d. A sort algorithm

Extra short questions

1. Four digital presentation tools in the book.
2. When spreadsheet > slides.
3. When infographic > report.
4. One rule for all four.

Extra long question

1. Same finding (70% of surveyed students want longer labs). Design: a spreadsheet cell, a slide headline, an infographic hook, and a report paragraph.

Extra answer key

MCQs: 1. A 2. A 3. A 4. A 5. A 6. A

Short

1. Spreadsheets, slides, infographics, reports.
2. When the audience must check the numbers or filter.
3. Awareness poster / social post — not a methods thesis.
4. Cite sources; label axes; do not crop axes to lie.

Long

1. Sheet: table+bar. Slide: one number, one chart, one ask. Infographic: 7 of 10 icon row.
Report: sample n, question wording, 70%, limit (one college).

6.7.1 – 6.7.4 Spreadsheets, Presentations, Infographics, Reports

6.7.3 Infographics

LEARN IT

An **infographic** (information + graphic) is a **visual way** of presenting information, facts or data so that people can understand the main idea **quickly and easily**. Tools such as **Canva, Power BI** or **Tableau** can be used. A good infographic should:

- **Simplify** complex concepts.
- **Highlight** the most important findings and results.
- Use **clear visuals, limited text** and **accurate** information.

6.7.4 Reports

LEARN IT

Reports are **structured documents** used to present collected data, findings, analysis and conclusions. Tools like **MS Word** or **Google Docs** are standard. A good report should:

- Start with a clear **title, introduction** and **purpose**.
- Present the collected data using **tables or charts**.
- Explain findings in **simple language**.
- Include **conclusions and recommendations** based on the data.

Optional: you may use AI tools carefully to improve wording, but **always review for accuracy**. AI can make mistakes or invent facts, and you are responsible for what you submit.

LEARN IT

Visual rule of thumb: a good chart or infographic can explain complex data **faster** than a long paragraph. However, visuals must always be **accurate** and based strictly on the **real data collected** – never exaggerated.

WORKED EXAMPLE

Which tool should you choose?

Situation	Best tool	Why
You must calculate the average marks of 40 students.	Spreadsheet	Formulas calculate instantly.
You will explain results to the class in assembly.	Presentation	Designed for a live audience.
You want a one-page poster for the notice board.	Infographic	Main result understood at a glance.
The principal wants the full study with method and recommendations.	Report	Detailed, structured document.

REMEMBER IT

Match tool to job: **Spreadsheet = Calculate**, **Presentation = Tell**, **Infographic = Show quickly**, **Report = Explain fully**.

COMMON MISTAKE

Mistake: Making a chart look more dramatic than the data – for example, starting the bar-chart axis at 18 instead of 0 so that 20 looks twice as big as 19. **Fix:** Start bar-chart axes at 0, label axes, and show real numbers only.

اردو میں سمجھیں

انفوگرافک معلومات کو تصویری انداز میں پیش کرنے کا طریقہ ہے تاکہ اصل بات جلدی سمجھ آ جائے؛ اس کے لیے کینوا (Canva) اور پاور بی آئی (Power BI) جیسے ٹولز استعمال ہوتے ہیں۔ رپورٹ ایک منظم دستاویز ہے جس میں عنوان، تعارف، ڈیٹا، نتائج، اختتامیہ اور سفارشات شامل ہوتی ہیں۔ تصاویر اور چارٹس ہمیشہ اصل ڈیٹا کے مطابق اور درست ہونے چاہئیں، مبالغہ آمیز نہیں۔

CHECK YOURSELF

1. Name two tools for making infographics.
2. What must you always do if you use AI to improve your report's wording?
3. What should a report end with?

Answers: 1. Any two: Canva, Power BI, Tableau. 2. Review it carefully for accuracy. 3. Conclusions and recommendations based on the data.

Extra board practice

These MCQs and questions are extra to the textbook. Close the notes, attempt them, then open the key.

MCQ 1. Spreadsheets are good for:

- a. Tables, calculations, simple charts
- b. 3D games
- c. NAND simulation
- d. OSI sniffing

MCQ 2. A presentation (slides) is for:

- a. Talking an audience through findings
- b. Storing the only copy of raw data
- c. Primary keys
- d. Kernel drivers

MCQ 3. An infographic is:

- a. A visual one-pager of facts
- b. A 40-page appendix
- c. A binary search
- d. A foreign key

MCQ 4. A report is:

- a. Structured writing with method, results, limits
- b. Only memes
- c. Only code
- d. Only ER

MCQ 5. Choose the tool by:

- a. Audience and purpose
- b. Whatever is newest
- c. Always infographic
- d. Always 80 slides

MCQ 6. Raw 2000-row sheets in a 5-minute talk are:

- a. A bad present
- b. Ideal
- c. Required by BIEK
- d. A sort algorithm

Extra short questions

1. Four digital presentation tools in the book.
2. When spreadsheet > slides.
3. When infographic > report.
4. One rule for all four.

Extra long question

1. Same finding (70% of surveyed students want longer labs). Design: a spreadsheet cell, a slide headline, an infographic hook, and a report paragraph.

Extra answer key

MCQs: 1. A 2. A 3. A 4. A 5. A 6. A

Short

1. Spreadsheets, slides, infographics, reports.
2. When the audience must check the numbers or filter.
3. Awareness poster / social post — not a methods thesis.
4. Cite sources; label axes; do not crop axes to lie.

Long

1. Sheet: table+bar. Slide: one number, one chart, one ask. Infographic: 7 of 10 icon row.
Report: sample n, question wording, 70%, limit (one college).

6.8 Case Study: Digital Inquiry Project ★ Golden

LEARN IT

This case study shows how to apply **all** the ideas of this chapter to solve a real-world problem from start to finish. An **inquiry** is an investigation to find the answer to a question.

Research question: "How does daily mobile phone use affect the study habits of students?"

Scenario: Students notice classmates using phones during free periods and while studying. Some say phones **help** them learn; others say phones **waste time**. The class must investigate whether phone use helps or harms study habits.

The digital inquiry workflow



The project is carried out in **12 steps** that fit into this five-stage workflow:

Workflow stage	Steps
Question & Search	1 Advanced searches · 2 Identify data types · 3 Select methods
Collect	4 Prepare the survey · 5 Choose the sample · 6 Collect primary data ethically · 7 Use secondary data
Analyse	8 Organise in a spreadsheet · 9 Create charts · 10 Analyse results
Create	11 Write the conclusion · 12 Recommendations → final digital artefact

REMEMBER IT

Workflow in five words: **Ask → Search → Collect → Analyse → Create**. Every research project you ever do will follow this pattern.

اردو میں سمجھیں

اس کیس اسٹڈی میں تحقیقی سوال یہ ہے: "روزانہ موبائل فون کا استعمال طلبہ کی پڑھائی کی عادات پر کیسے اثر ڈالتا ہے؟" تحقیق پانچ مراحل میں ہوتی ہے: سوال، تلاش، ڈیٹا جمع کرنا، تجزیہ اور آخری پروڈکٹ بنانا۔ یہ کام بارہ قدموں میں مکمل کیا جاتا ہے۔

EXAM TIP

Expect a long question: "Describe the steps of a digital inquiry project using the example of mobile phone use." Write the research question first, then the 12 steps with one or two lines each, and finish with the artefact. Learn the five-stage workflow as your skeleton.

CHECK YOURSELF

1. What is the research question in the case study?
2. Name the five stages of the digital inquiry workflow.

Answers: 1. "How does daily mobile phone use affect the study habits of students?" 2. Question, Search, Collect, Analyse, Create.

Extra board practice

These MCQs and questions are extra to the textbook. Close the notes, attempt them, then open the key.

MCQ 1. The digital inquiry project applies:

- a. All of chapter 6 in order
- b. Only careers
- c. Only NAND
- d. Only OSI

MCQ 2. Advanced search helps you:

- a. Narrow secondary sources (quotes, site, date)
- b. Hack Wi-Fi
- c. Draw K-maps
- d. Sort in $O(1)$

MCQ 3. Primary collection in the project is often:

- a. A class survey
- b. A CPU design
- c. A TCP stack
- d. A waterfall contract

MCQ 4. You organise results in a:

- a. Spreadsheet + charts
- b. Random WhatsApp dump only
- c. NAND netlist
- d. Figma only

MCQ 5. The artefact is:

- a. The final digital product that answers the question
- b. The first brainstorm sticky
- c. The career list
- d. The teacher's attendance

MCQ 6. A conclusion should:

- a. Answer the question and admit limits
- b. Ignore the data
- c. Name and shame students
- d. Invent $n=1$ million

Extra short questions

1. List the project steps in order (grouped as in the book).
2. What is a digital artefact here?
3. Why include secondary if you surveyed?
4. One sentence of a good limitation.

Extra long question

1. Outline a 12-step inquiry: "Does extra screen time lower concentration in our class?" Name tools and the artefact. ★ Golden.

Extra answer key

MCQs: 1. A 2. A 3. A 4. A 5. A 6. A

Short

1. Search/method; survey/primary; secondary+sheet; analyse/conclude; artefact.
2. Slides, infographic, short video, site — that answers the inquiry question.
3. To compare your small sample with a published claim.
4. $n=40$ in one college, so we cannot speak for all Sindh.

Long

1. Steps 1–3 search; 4–6 survey; 7–9 PBS/article + sheet; 10–12 charts, conclusion, artefact (infographic+spreadsheet). Ethics. One expected chart: hours vs self-reported focus.

Step 1 – 3: Advanced Search & Methodology ★ Golden

The **methodology** is the plan of methods a researcher uses to answer the question.

Step 1: Perform advanced searches

LEARN IT

Students first search for **background information** using search engines (such as Google or Bing). They use **advanced search operators** – special symbols or words typed into the search box – to **narrow down** results and find **reliable secondary data**.

Suggested query	Operator used	What it does
"mobile phone use" "study habits" students	" " (quotes)	Finds the exact phrases "mobile phone use" and "study habits", not the words scattered randomly.
"academic performance" filetype:pdf	filetype:	Finds only PDF documents (often official reports and research papers).
digital distraction site:.edu	site:	Searches only educational websites (addresses ending in .edu).

Other useful operators (background): - (minus) excludes a word, e.g. phone addiction - game; OR finds either word, e.g. smartphone OR mobile; site:.gov.pk searches only Pakistani government websites.

LEARN IT

Source evaluation checklist. Before trusting a source, ask:

- | | |
|---|--|
| ☐ Is the author/organisation clearly mentioned? | ☐ Is the information recent ? |
| ☐ Is the website reliable (.edu, .gov, research-based)? | ☐ Does the source provide facts/references , or is it biased (one-sided, trying to sell something)? |

WORKED EXAMPLE

Building a search query step by step. You want official PDF reports about screen time of students on Pakistani government websites.

Step 1 – Plain search: `screen time students` → millions of results, many blogs and adverts.

Step 2 – Exact phrase: `"screen time" students` → results must contain the exact phrase.

Step 3 – Only PDFs: `"screen time" students filetype:pdf` → mostly reports and papers.

Step 4 – Only government sites: `"screen time" students filetype:pdf site:.gov.pk` → a small number of highly relevant, reliable results.

Step 5 – Evaluate: open a result and check the checklist: author named? year recent? references given?

Step 2: Identify data types

LEARN IT

Students decide what kind of **primary data** they will collect:

- **Quantitative:** number of hours used daily; percentage who feel distracted.
- **Qualitative:** opinions on why they get distracted; suggestions for better use.

Step 3: Select data-collection methods

LEARN IT

Because they need to ask the **same questions to many students**, a **survey/questionnaire** (via Google Forms or on paper) is the **most efficient** method. Interviews would take too long for so many people, and observation could not reveal opinions.

REMEMBER IT

Three operators, three jobs: **" "** = **exact**, **filetype:** = **format**, **site:** = **where**. Note: there is **no space** after the colon (`site:.edu`, not `site: .edu`).

COMMON MISTAKE

Mistake: Assuming the first search result is the most reliable. Search results can be adverts or popular blogs. **Fix:** Always apply the source evaluation checklist. A website ending in `.edu` or `.gov` is usually more reliable, but you should still check the date and author.

پہلے قدم میں طلبہ سرچ انجن پر جدید سرچ آپریٹرز (Advanced Search Operators) استعمال کرتے ہیں: کوئیشن مارکس (" ") بالکل وہی جملہ تلاش کرتے ہیں، filetype:pdf صرف پی ڈی ایف فائلیں اور site:.edu صرف تعلیمی ویب سائٹس دکھاتا ہے۔ ہر ذریعے کو مصنف، تاریخ، قابل اعتماد ویب سائٹ اور حوالوں کی بنیاد پر پرکھا جاتا ہے۔ پھر ڈیٹا کی اقسام طے کی جاتی ہیں اور بہت سے طلبہ سے سوال کرنے کے لیے سروے کو بہترین طریقہ چنا جاتا ہے۔

EXAM TIP

"Explain any three advanced search operators with examples" is a likely short question. For each, write the operator, an example query, and what it does. Also learn the four-point source evaluation checklist.

CHECK YOURSELF

1. Write a query that finds the exact phrase "online learning" only on educational websites.
2. Why was a survey chosen in Step 3?
3. Give one quantitative and one qualitative data item from Step 2.

Answers: 1. "online learning" site:.edu . 2. Because the same questions had to be asked to many students, and a survey is the most efficient way to do that. 3. Quantitative: hours used daily (or % who feel distracted). Qualitative: opinions on why they get distracted (or suggestions for better use).

Extra board practice

These MCQs and questions are extra to the textbook. Close the notes, attempt them, then open the key.

MCQ 1. The digital inquiry project applies:

- a. All of chapter 6 in order
- b. Only careers
- c. Only NAND
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- a. Narrow secondary sources (quotes, site, date)
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- a. The final digital product that answers the question
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MCQ 6. A conclusion should:

- a. Answer the question and admit limits
- b. Ignore the data
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- d. Invent $n=1$ million

Extra short questions

1. List the project steps in order (grouped as in the book).
2. What is a digital artefact here?
3. Why include secondary if you surveyed?
4. One sentence of a good limitation.

Extra long question

1. Outline a 12-step inquiry: "Does extra screen time lower concentration in our class?" Name tools and the artefact. ★ Golden.

Extra answer key

MCQs: 1. A 2. A 3. A 4. A 5. A 6. A

Short

1. Search/method; survey/primary; secondary+sheet; analyse/conclude; artefact.
2. Slides, infographic, short video, site — that answers the inquiry question.
3. To compare your small sample with a published claim.
4. $n=40$ in one college, so we cannot speak for all Sindh.

Long

1. Steps 1–3 search; 4–6 survey; 7–9 PBS/article + sheet; 10–12 charts, conclusion, artefact (infographic+spreadsheet). Ethics. One expected chart: hours vs self-reported focus.

Step 4 – 6: Survey Prep & Primary Data Collection ★ Golden

Step 4: Prepare the survey

LEARN IT

Design **clear, unbiased** questions. Each question is linked to the type of data it produces:

Question	Type of data
How many hours do you use your phone daily?	Quantitative
Which activity do you use your phone for the most?	Quantitative (categories)
How does your phone help you in studies?	Qualitative
Suggest one way to use phones responsibly.	Qualitative

Step 5: Choose the sample

LEARN IT

The **sample** is the group of people who provide the data. It is usually impossible to ask everyone (the whole **population**), so a smaller group is chosen. The students decide to survey **40 Class XI students from different sections** to get a **representative** mix of habits. "Representative" means the sample looks like the whole group – it includes different kinds of students, not just your friends.

Step 6: Collect primary data (ethically)

LEARN IT

Ethics means doing what is morally right. When conducting the survey, students must follow strict ethical guidelines:

- Ask **permission** before collecting responses (**consent**).
- Avoid collecting **sensitive personal information**.
- Do **not record names** unless absolutely necessary (**anonymity**).
- Keep the responses **private**.
- Use the data **only for educational purposes**.

WORKED EXAMPLE

Why "different sections" matters. Suppose the group surveyed only the 40 students of the Pre-Engineering section who sit in the front rows.

Problem: these students might study more than average, so the results would show less phone distraction than is true for the whole class.

Fix: pick students from every section (e.g. 10 each from Pre-Medical, Pre-Engineering, Computer Science and Commerce). Now the sample better represents all Class XI students.

Ethical opening text for the form: "This survey is for a class project on phone use and study habits. It is anonymous – do not write your name. Your answers will be kept private and used only for this project. Do you agree to take part? O Yes O No"

REMEMBER IT

Ethics in five words: **Consent, No-Sensitive, Anonymous, Private, Educational** – "**C-NAPE**". Think: "Cover the person's nape (neck) – protect them."

COMMON MISTAKE

Mistake: Thinking a bigger sample is always enough. A sample of 200 students all from one section is still biased. **Fix:** A sample must be **representative**, not just large.

اردو میں سمجھیں

چوتھے قدم میں واضح اور غیر جانبدار سوالات تیار کیے جاتے ہیں۔ پانچویں قدم میں نمونہ (Sample) چنا جاتا ہے: مختلف سیکشنز کے چالیس طلبہ تاکہ نتائج پوری جماعت کی نمائندگی کریں۔ چھٹے قدم میں اخلاقی اصولوں کے ساتھ ڈیٹا جمع کیا جاتا ہے: اجازت لیں، حساس معلومات نہ لیں، نام نہ لکھیں (گمنامی)، جوابات کو راز میں رکھیں اور صرف تعلیمی مقصد کے لیے استعمال کریں۔

EXAM TIP

Questions often ask: "What ethical guidelines should be followed when collecting data?" List all five points. Use the words **consent** and **anonymity** – examiners look for them. You may also be asked to classify the four survey questions by data type.

CHECK YOURSELF

1. What is a sample?
2. What does anonymity mean in a survey?
3. What type of data does "Suggest one way to use phones responsibly" give?

Answers: 1. The group of people who provide the data. 2. Not recording the names of participants (unless absolutely necessary). 3. Qualitative.

Extra board practice

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- c. Only NAND
- d. Only OSI

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- a. Narrow secondary sources (quotes, site, date)
- b. Hack Wi-Fi
- c. Draw K-maps
- d. Sort in $O(1)$

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MCQ 4. You organise results in a:

- a. Spreadsheet + charts
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- c. NAND netlist
- d. Figma only

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- a. The final digital product that answers the question
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- c. The career list
- d. The teacher's attendance

MCQ 6. A conclusion should:

- a. Answer the question and admit limits
- b. Ignore the data
- c. Name and shame students
- d. Invent $n=1$ million

Extra short questions

1. List the project steps in order (grouped as in the book).
2. What is a digital artefact here?
3. Why include secondary if you surveyed?
4. One sentence of a good limitation.

Extra long question

1. Outline a 12-step inquiry: "Does extra screen time lower concentration in our class?" Name tools and the artefact. ★ Golden.

Extra answer key

MCQs: 1. A 2. A 3. A 4. A 5. A 6. A

Short

1. Search/method; survey/primary; secondary+sheet; analyse/conclude; artefact.
2. Slides, infographic, short video, site — that answers the inquiry question.
3. To compare your small sample with a published claim.
4. $n=40$ in one college, so we cannot speak for all Sindh.

Long

1. Steps 1–3 search; 4–6 survey; 7–9 PBS/article + sheet; 10–12 charts, conclusion, artefact (infographic+spreadsheet). Ethics. One expected chart: hours vs self-reported focus.

Step 7 – 9: Secondary Data & Spreadsheet Organization ★ Golden

Step 7: Use secondary data

LEARN IT

Students include a reliable online article stating that "**excessive screen time lowers concentration.**" They will **compare** this secondary data with their own primary survey results. If both agree, the conclusion is stronger. If they disagree, the students must think about why (for example, a different age group or country).

Step 8: Organise data in a spreadsheet

LEARN IT

Responses are entered into **MS Excel** or **Google Sheets** to **sort, calculate totals** and **find averages**. Each row is one student; each column is one question.

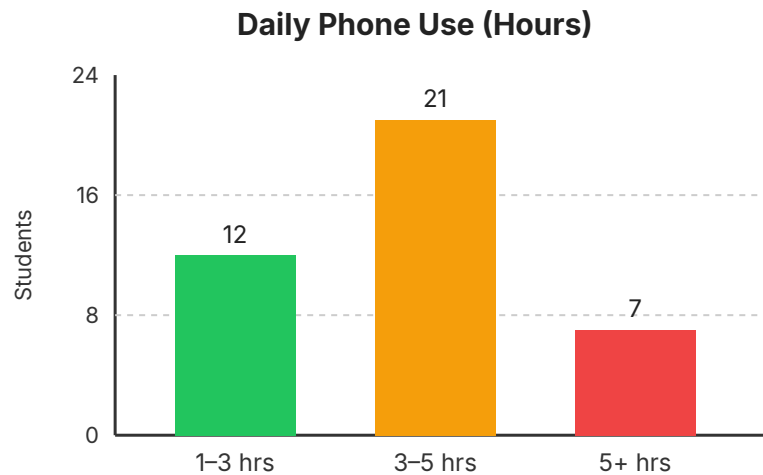
Student	Hours	Study Use	Main Activity	Distracted?
1	3	Yes	Lectures	Sometimes
2	5	No	Social Media	Yes
3	2	Yes	Notes	No

(The full sheet has 40 rows, one per student. The formulas for this table are worked out in topic 6.7.1.)

Step 9: Create charts and graphs

LEARN IT

The spreadsheet data is converted into **visuals** to make it easy to digest. In the source, the "Daily Phone Use (Hours)" bar chart groups the 40 students into three bars: 1–3 hrs, 3–5 hrs and 5+ hrs, with the 3–5 hrs bar the tallest.



Bar chart of daily phone use for 40 students. The counts (12, 21, 7) are estimated from the bar heights in the source chart.

WORKED EXAMPLE

Making the chart in Google Sheets.

Step 1 – Count each group. If the hours are in B2:B41, use `=COUNTIF(B2:B41,"<3")` for under 3 hrs, `=COUNTIFS(B2:B41,"≥3",B2:B41,"<5")` for 3 to under 5 hrs, and `=COUNTIF(B2:B41,"≥5")` for 5 hrs or more. (This gives non-overlapping groups.)

Step 2 – Check the total. $12 + 21 + 7 = 40$ ✓ – every student is counted exactly once.

Step 3 – Percentages. $12 \div 40 = 30\%$; $21 \div 40 = 52.5\%$; $7 \div 40 = 17.5\%$. Total = 100% ✓.

Step 4 – Insert chart. Select the group labels and counts → Insert → Chart → choose *Column chart*. Add the title "Daily Phone Use (Hours)" and label the axes.

Step 5 – Choose the right chart. Bar/column chart to compare groups; pie chart to show parts of a whole (e.g. % in each group); line chart to show change over time.

REMEMBER IT

Chart chooser: **Bar = Compare**, **Pie = Parts of a whole**, **Line = Change over time**.

COMMON MISTAKE

Mistake: Using overlapping groups like "1–3" and "3–5": a student who uses the phone for exactly 3 hours could be counted twice or put in the wrong bar. **Fix:** Define groups clearly, e.g. "under 3", "3 to under 5", "5 or more", and check that the group counts add up to the sample size.

ساتویں قدم میں ایک قابلِ اعتماد آن لائن مضمون کا ثانوی ڈیٹا ("زیادہ اسکرین ٹائم توجہ کم کرتا ہے") اپنے سروے کے بنیادی ڈیٹا سے موازنہ کرنے کے لیے لیا جاتا ہے۔ آٹھویں قدم میں جوابات ایکسل یا گوگل شیٹس میں درج کر کے ترتیب، مجموعہ اور اوسط نکالی جاتی ہے۔ نویں قدم میں اسپریڈشیٹ کے ڈیٹا سے چارٹ اور گراف بنائے جاتے ہیں تاکہ نتائج آسانی سے سمجھ آئیں۔

EXAM TIP

You may get the spreadsheet table and be asked to "write formulas to find the total and average hours" or "count students who use the phone for study". Write the full formula with =, the correct range, and the result.

CHECK YOURSELF

1. What is the secondary data used in Step 7?
2. Which formula counts how many students answered "Yes" in column C, rows 2 to 41?
3. Which type of chart is best for comparing the three hour-groups?

Answers: 1. A reliable online article stating "excessive screen time lowers concentration". 2. =COUNTIF(C2:C41, "Yes") . 3. A bar (column) chart.

Extra board practice

These MCQs and questions are extra to the textbook. Close the notes, attempt them, then open the key.

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Extra short questions

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1. Search/method; survey/primary; secondary+sheet; analyse/conclude; artefact.
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Long

1. Steps 1–3 search; 4–6 survey; 7–9 PBS/article + sheet; 10–12 charts, conclusion, artefact (infographic+spreadsheet). Ethics. One expected chart: hours vs self-reported focus.

Step 10 – 12: Data Analysis, Conclusion & Artefact ★ Golden

Step 10: Analyse the results

LEARN IT

To **analyse** means to study the data carefully to find **patterns** and **trends**. By looking at the spreadsheet and charts, the group identifies:

- **Most students** (the majority) use phones for **3–5 hours daily**.
- **Half** use it for **educational purposes** (lectures/notes).
- **A clear trend:** students who use phones mostly for **entertainment** report much **higher distraction** levels than those who use them for study.

Step 11: Write the conclusion

LEARN IT

A **conclusion** answers the research question using the evidence:

"The research shows mobile phones have both positive and negative effects. They provide access to learning materials. However, excessive use of social media and gaming reduces concentration. Therefore, phones should not be banned, but managed wisely."

Step 12: Recommendations

LEARN IT

Recommendations are practical suggestions based on the data. The students recommend:

- Turn off **social media notifications** while studying.
- Set **daily screen-time limits**.
- Take **physical breaks** from screens to protect health.

WORKED EXAMPLE

From numbers to a finding. Using the estimated chart counts (12, 21, 7 out of 40):

Step 1: $21 \div 40 = 52.5\%$. Since $52.5\% > 50\%$, the statement "most students use phones for 3–5 hours" is supported.

Step 2: "Half use it for educational purposes" means about 20 of the 40 students chose Lectures or Notes as their main activity ($20 \div 40 = 50\%$). Formula:

=COUNTIF(D2:D41, "Lectures")+COUNTIF(D2:D41, "Notes") .

Step 3 – Compare groups: filter the sheet to show only Social Media/Games users and count how many said "Yes" to distracted; then do the same for Lectures/Notes users. If the entertainment group has a much higher share of "Yes", that is the trend.

Step 4 – Link to secondary data: the article said excessive screen time lowers concentration; the survey agrees, so the conclusion is stronger.

REMEMBER IT

Conclusion formula: "**The data shows ... However ... Therefore ...**" – state the main finding, mention the other side, then give the overall answer.

COMMON MISTAKE

Mistake: Writing a conclusion that is not based on the data, e.g. "Phones should be banned" when the data showed phones also help learning. **Fix:** Every sentence in the conclusion and recommendations must be supported by your findings. Also, a trend between two things (entertainment use and distraction) shows a *link*; be careful before claiming one definitely *causes* the other.

اردو میں سمجھیں

دسویں قدم میں چارٹس اور اسپریڈشیٹ سے رجحانات تلاش کیے جاتے ہیں: زیادہ تر طلبہ روزانہ تین سے پانچ گھنٹے فون استعمال کرتے ہیں، نصف اسے تعلیم کے لیے استعمال کرتے ہیں، اور تفریح کے لیے استعمال کرنے والے زیادہ توجہ بٹنے کی شکایت کرتے ہیں۔ گیارہویں قدم میں نتیجہ لکھا جاتا ہے کہ فون پر پابندی نہیں بلکہ اس کا دانشمندانہ استعمال ہونا چاہیے۔ بارہویں قدم میں سفارشات دی جاتی ہیں: پڑھائی کے دوران نوٹیفکیشن بند کریں، اسکرین ٹائم کی حد مقرر کریں اور وقفے لیں۔

EXAM TIP

You may be asked to "write a conclusion and two recommendations" for given data. Use the "shows ... however ... therefore" pattern and make sure each recommendation follows from a finding.

CHECK YOURSELF

1. According to Step 10, which group reported higher distraction?
2. Should phones be banned according to the conclusion?
3. Give two of the students' recommendations.

Answers: 1. Students who use phones mostly for entertainment. 2. No – they should not be banned but managed wisely. 3. Any two: turn off social media notifications while studying; set daily screen-time limits; take physical breaks from screens.

LEARN IT

A **digital artefact** is a final product created using digital technologies. Students must create **one final artefact** that answers the original research question. It is **not just a decorated file**; it must present **evidence** and **conclusions**.

Artefact option	Suitable digital tools	Main purpose
Digital Report	MS Word, Google Docs	Explain the full study in a structured text document.
Presentation	PowerPoint, Canva, Slides	Share key findings with a live audience.
Infographic	Canva, Adobe Express	Show the main results visually and quickly on one page.
Dashboard	Power BI, Tableau	Explore data through interactive charts and filters.

A **dashboard** is an interactive screen of charts; the viewer can click filters (e.g. "show only Commerce students") and the charts update.

Standard artefact structure

1. Title
2. Research Question
3. Introduction
4. Method of Data Collection
5. Primary Data Findings
6. Secondary Data Summary
7. Charts and Graphs
8. Conclusion
9. Recommendations
10. References

WORKED EXAMPLE

Planning a 10-slide presentation artefact for the phone study (one slide per structure item):

Slide 1 Title: "Phones and Study Habits – Class XI Survey". **Slide 2** Research question. **Slide 3** Introduction: why the class investigated this. **Slide 4** Method: Google Form survey, 40 students, different sections, anonymous with consent. **Slide 5** Primary findings: 52.5% use phones 3–5 hours; half use for study. **Slide 6** Secondary data: article on screen time and concentration. **Slide 7** The bar chart of daily phone use. **Slide 8** Conclusion: "manage wisely, don't ban". **Slide 9** Three recommendations. **Slide 10** References: the article's title, author, website and date.

REMEMBER IT

The structure follows the inquiry itself: **What** (title, question, intro) → **How** (method) → **Found** (primary, secondary, charts) → **So what** (conclusion, recommendations) → **Credit** (references).

COMMON MISTAKE

Mistake: Spending all your time on colours and animations and forgetting evidence. **Fix:** An artefact is judged on whether it answers the research question with evidence. Also never skip **References** – listing your secondary sources is honest and avoids plagiarism.

اردو میں سمجھیں

ڈیجیٹل آرٹیفیکٹ (Digital Artefact) ڈیجیٹل ٹیکنالوجی سے بنائی گئی آخری پروڈکٹ ہے جو تحقیقی سوال کا جواب ثبوت اور نتائج کے ساتھ پیش کرتی ہے؛ یہ صرف سچی ہوئی فائل نہیں۔ اس کی چار صورتیں ہیں: رپورٹ، پریزنٹیشن، انفوگرافک اور ڈیش بورڈ۔ معیاری ساخت میں عنوان، تحقیقی سوال، تعارف، طریقہ کار، بنیادی و ثانوی ڈیٹا، چارٹس، نتیجہ، سفارشات اور حوالہ جات شامل ہیں۔

EXAM TIP

Learn the four artefact options with tools and purpose (a table question), and the ten-part structure in order. "What is a digital artefact?" – stress that it must present **evidence and conclusions**, not just decoration.

CHECK YOURSELF

1. Which artefact lets viewers explore data with interactive filters?
2. What is the last item in the standard artefact structure?
3. Name two tools for making an infographic artefact.

Answers: 1. A dashboard (Power BI, Tableau). 2. References. 3. Canva and Adobe Express.

Extra board practice

These MCQs and questions are extra to the textbook. Close the notes, attempt them, then open the key.

MCQ 1. The digital inquiry project applies:

- a. All of chapter 6 in order
- b. Only careers
- c. Only NAND
- d. Only OSI

MCQ 2. Advanced search helps you:

- a. Narrow secondary sources (quotes, site, date)
- b. Hack Wi-Fi
- c. Draw K-maps
- d. Sort in $O(1)$

MCQ 3. Primary collection in the project is often:

- a. A class survey
- b. A CPU design
- c. A TCP stack
- d. A waterfall contract

MCQ 4. You organise results in a:

- a. Spreadsheet + charts
- b. Random WhatsApp dump only
- c. NAND netlist
- d. Figma only

MCQ 5. The artefact is:

- a. The final digital product that answers the question
- b. The first brainstorm sticky
- c. The career list
- d. The teacher's attendance

MCQ 6. A conclusion should:

- a. Answer the question and admit limits
- b. Ignore the data
- c. Name and shame students
- d. Invent $n=1$ million

Extra short questions

1. List the project steps in order (grouped as in the book).
2. What is a digital artefact here?
3. Why include secondary if you surveyed?
4. One sentence of a good limitation.

Extra long question

1. Outline a 12-step inquiry: "Does extra screen time lower concentration in our class?" Name tools and the artefact. ★ Golden.

Extra answer key

MCQs: 1. A 2. A 3. A 4. A 5. A 6. A

Short

1. Search/method; survey/primary; secondary+sheet; analyse/conclude; artefact.
2. Slides, infographic, short video, site — that answers the inquiry question.
3. To compare your small sample with a published claim.
4. $n=40$ in one college, so we cannot speak for all Sindh.

Long

1. Steps 1–3 search; 4–6 survey; 7–9 PBS/article + sheet; 10–12 charts, conclusion, artefact (infographic+spreadsheet). Ethics. One expected chart: hours vs self-reported focus.

Digital Literacy

14 single-page lessons · 10 Golden · read, close, say aloud, then sit the exam

Chapter in one look

- **Digital literacy** is the ability to understand and use digital devices, the internet and online tools safely, effectively and responsibly – not just using devices, but finding, evaluating and creating information.
- In the **Information Age**, access to technology is not enough; digital literacy needs technical, cognitive and social abilities.
- **Qualitative data** is descriptive and non-numerical (opinions, reasons, behaviour) and explains why/how.
- **Quantitative data** is numerical, countable or measurable (hours, scores, percentages) and is used for comparing and charting.
- The five **data-collection strategies** are interviews, surveys, prototypes, observation and simulations.
- **Interviews** are planned conversations giving mostly qualitative data; **surveys** ask the same questions to many people, on paper or with Google/Microsoft Forms.
- Good survey questions are clear, short, unbiased and ask about one thing; avoid leading, double-barrelled, vague and overlapping questions.
- A **prototype** is an early version tested for feedback; **observation** records what happens without asking questions; a **simulation** is a computer model used when real experiments are costly, risky or difficult.
- **Primary data** is collected first-hand (high control); **secondary data** was collected by others and reused (limited control).
- Designing a data-collection approach has eight steps: question, data type, method, sample, tools, collect, store with backup, and ethics (consent is mandatory).
- **Spreadsheets** organise and calculate (e.g. `=SUM()`, `=AVERAGE()`, `=COUNTIF()`); **presentations** deliver to an audience; **infographics** give quick visual summaries; **reports** give detailed structure.
- Visuals must always be accurate and based strictly on real data, never exaggerated.
- **Advanced search operators** such as `" "`, `filetype:pdf` and `site:.edu` help find reliable sources efficiently; sources must be evaluated for author, date, reliability and bias.
- **Ethics**: get consent, avoid sensitive information, keep anonymity, keep responses private, and use data only for educational purposes.
- The **digital inquiry project** follows Question → Search → Collect → Analyse → Create in 12 steps.
- A **digital artefact** (report, presentation, infographic or dashboard) answers the research question using evidence and visuals, following a ten-part structure ending in references.

Key terms

Term	Meaning	اردو
Digital literacy	Ability to understand and use digital devices, the internet and online tools safely, effectively and responsibly.	ڈیجیٹل خواندگی
Information Age	Period in which vast amounts of data are produced, shared and accessed instantly.	معلوماتی دور
Data	Facts, numbers, words or observations collected for study.	مواد / ڈیٹا

Qualitative data	Descriptive, non-numerical data about thoughts, feelings and experiences.	کیفیتی ڈیٹا
Quantitative data	Numerical data that can be counted or measured.	مقداری ڈیٹا
Interview	Planned conversation to gather information directly from a person or small group.	انٹرویو / بالمشافہ گفتگو
Survey / Questionnaire	Same set of questions asked to many people.	سروے / سوالنامہ
Prototype	Simple early version of a product used to test an idea and get feedback.	ابتدائی نمونہ
Observation	Watching and recording what happens without asking questions.	مشاہدہ
Simulation	Computer-based model of a real situation in which variables can be changed.	نقلی ماڈل (سمیولیشن)
Variable	A factor that can be changed or measured.	متغیر
Primary data	Data collected first-hand by the researcher for a specific purpose.	بنیادی ڈیٹا

CHAPTER 6 EXAM • CLOSED BOOK

Digital Literacy

75 marks • 1 hour 15 minutes • no notes • no phones

CHAPTER EXAM RULES

Instructions. Sit this exam after Lesson 14 and the recap. **Section A** — all 15 MCQs (1 mark). **Section B** — all 10 short questions (3 marks). **Section C** — attempt any **three** long questions (10 marks). At least one Section C answer must be a ★ Golden topic. Write in your notebook. Do not open the key until time is up.

Marks. 15 + 30 + 30 = 75. A 5-mark / 10-mark answer needs Define • Explain • Example • Diagram • Working.

سبق اور خلاصے کے بعد یہ امتحان بغیر نوٹس کے حل کریں۔ وقت پورا ہونے پر کلید کھولیں۔

Section A — Multiple choice (15 × 1 = 15)

1. Digital literacy is best described as the ability to:
 - a. type quickly on a keyboard
 - b. understand and use digital devices, the internet and online tools safely, effectively and responsibly
 - c. repair computer hardware
 - d. write computer programs
2. Which of the following is qualitative data?
 - a. Marks obtained in a test
 - b. Number of hours spent online
 - c. Students' opinions about online classes
 - d. Percentage of students using the internet
3. Which method is most suitable when the same questions must be asked to many people?
 - a. Survey
 - b. Interview
 - c. Prototype
 - d. Simulation
4. Simulations are especially useful when:
 - a. you want people's opinions
 - b. the data already exists in a report
 - c. you want to watch people without asking questions
 - d. real experiments are costly, risky or difficult
5. To find out how many cars cross an intersection at 8 AM, the best method is:
 - a. Interview
 - b. Observation
 - c. Survey
 - d. Prototype
6. Which of these is primary data?
 - a. Responses to a Google Form you created for your project
 - b. An official government report
 - c. A published research article
 - d. Web statistics from a website
7. A researcher has limited control over:
 - a. their own interviews
 - b. their own survey questions
 - c. their own observations
 - d. secondary data
8. The first step in designing a data-collection approach is to:
 - a. collect the data
 - b. choose the sample
 - c. define the research question clearly
 - d. store the data with a backup
9. Which spreadsheet formula finds the mean of cells B2 to B6?
 - a. `=AVERAGE(B2:B6)`
 - b. `=SUM(B2:B6)`
 - c. `=COUNT(B2:B6)`
 - d. `=MAX(B2:B6)`

- .0. The search query `digital distraction site:.edu` will:
- a. find only PDF files
 - b. search only educational websites
 - c. find the exact phrase "digital distraction" only
 - d. exclude educational websites
- .1. The operator `filetype:pdf` is used to:
- a. search for an exact phrase
 - b. search only government websites
 - c. convert a web page into PDF
 - d. find only PDF documents
- .2. "Do not record names unless absolutely necessary" describes the principle of:
- a. consent
 - b. accuracy
 - c. anonymity
 - d. sampling
- .3. In the case study, how many Class XI students were surveyed?
- a. 20
 - b. 40
 - c. 50
 - d. 100
- .4. Which tools are suggested for a dashboard artefact?
- a. Power BI, Tableau
 - b. MS Word, Google Docs
 - c. PowerPoint, Google Slides
 - d. Google Forms, Microsoft Forms
- .5. A prototype is:
- a. the final finished product
 - b. a published report
 - c. a type of survey question
 - d. a simple early version of a product used to test an idea

Section B — Short questions (10 × 3 = 30)

1. Define digital literacy. Name the three kinds of ability it involves.
2. Why is simply having access to technology not enough in the Information Age?
3. Differentiate between qualitative and quantitative data, giving two examples of each.
4. How is an interview different from a survey?
5. What is a prototype, and how does it produce useful data?
6. When is a simulation used? Give one example.
7. Compare primary and secondary data in terms of purpose and control.
8. List any four ethical guidelines to follow when collecting primary data.
9. Explain three advanced search operators with an example of each.
- .0. What is a digital artefact? Name its four options with one suitable tool each.

Section C — Long questions (attempt any 3 × 10 = 30)

1. Explain the eight steps of designing a data-collection approach. Apply each step to the research question: "How much water is wasted at the college taps during break time, and why?"
2. **Design a survey for** finding out how Class XI students in your college travel to college and how long the journey takes. Write a consent statement and at least six questions (mixing closed, number, rating and open-ended types), state the data type of each question, describe your sample, and explain how you will keep the data ethical.
3. Ten students recorded their daily phone use (hours) in cells B2 to B11: 2, 4, 3, 5, 6, 4, 3, 4, 1, 4. (a) Write the spreadsheet formulas and give the results for the total, mean, median, mode, maximum and minimum. (b) Write a formula to count students using the phone for more than 4 hours, and find the percentage. (c) Group the data into "1–2 hrs", "3–4 hrs" and "5+ hrs" and name a suitable chart.
4. Describe all 12 steps of the Digital Inquiry Project on the question "How does daily mobile phone use affect the study habits of students?", including the final digital artefact and its structure.
5. **Design a survey for** investigating "How do Class XI students use AI tools for homework?" Then explain which secondary sources you would search for (with advanced search queries), and which artefact you would create to present your findings to the principal, with reasons.

Answer key. Finish the Chapter Exam first, then mark. Do not peek while the clock is running.

پہلے امتحان مکمل کریں، پھر اس کلید سے نمبر لگائیں۔

Check yourself — answers

- 6.1 & 6.2 Introduction to Digital Literacy.** Answers: 1. The ability to understand and use digital devices, the internet and online tools in a safe, effective and responsible manner. 2. Find information, communicate, produce digital content, and solve problems through technology.
- 6.3 Types of Data (Qualitative vs Quantitative).** Answers: 1. Quantitative (measurable number). 2. Qualitative. 3. Qualitative (an opinion in words).
- 6.4 Data-Collection Strategies.** Answers: 1. Interviews, surveys, prototypes, observation, simulations. 2. New information that you gather yourself, rather than information that already exists.
- 6.4.1 – 6.4.2 Interviews and Surveys.** Answers: 1. Google Forms and Microsoft Forms. 2. It is double-barrelled (two subjects in one question); ask about each subject separately. 3. Usually qualitative.
- 6.4.3 – 6.4.5 Prototypes, Observation, and Simulations.** Answers: 1. Because the real experiment may be costly, risky or difficult. 2. Yes – for example, counting the number of cars passing a point. 3. The feedback users give after trying the prototype.
- 6.5 Primary and Secondary Data.** Answers: 1. Primary data. 2. Secondary (someone else collected it). 3. Any two: interviews, surveys, observations, experiments.
- 6.6 Designing a Data-Collection Approach.** Answers: 1. Define the research question clearly. 2. Step 8 – follow ethical practice and respect privacy. 3. Digitally, in a spreadsheet, with a backup copy.
- 6.7 Presenting Data Using Digital Tools.** Answers: 1. `=AVERAGE(D2:D20)`. 2. MS Excel, Google Sheets, Apple Numbers. 3. Any two: start with a clear title and research question; use short bullet points; include tables/charts/key findings; keep design simple and readable.
- 6.7.1 – 6.7.4 Spreadsheets, Presentations, Infographics, Reports.** Answers: 1. Any two: Canva, Power BI, Tableau. 2. Review it carefully for accuracy. 3. Conclusions and recommendations based on the data.
- 6.8 Case Study: Digital Inquiry Project.** Answers: 1. "How does daily mobile phone use affect the study habits of students?" 2. Question, Search, Collect, Analyse, Create.
- Step 1 – 3: Advanced Search & Methodology.** Answers: 1. "online learning" site:.edu. 2. Because the same questions had to be asked to many students, and a survey is the most efficient way to do that. 3. Quantitative: hours used daily (or % who feel distracted). Qualitative: opinions on why they get distracted (or suggestions for better use).
- Step 4 – 6: Survey Prep & Primary Data Collection.** Answers: 1. The group of people who provide the data. 2. Not recording the names of participants (unless absolutely necessary). 3. Qualitative.
- Step 7 – 9: Secondary Data & Spreadsheet Organization.** Answers: 1. A reliable online article stating "excessive screen time lowers concentration". 2. `=COUNTIF(C2:C41,"Yes")`. 3. A bar (column) chart.
- Step 10 – 12: Data Analysis, Conclusion & Artefact.** Answers: 1. Students who use phones mostly for entertainment. 2. No – they should not be banned but managed wisely. 3. Any two: turn off social media notifications while studying; set daily screen-time limits; take physical breaks from screens.

Answer key

Multiple choice

- | | | | | |
|------|------|------|-------|-------|
| 1. b | 4. d | 7. d | 10. b | 13. b |
| 2. c | 5. b | 8. c | 11. d | 14. a |
| 3. a | 6. a | 9. a | 12. c | 15. d |

Short questions — model answers

1. Digital literacy is the ability to understand and use digital devices, the internet and online tools in a safe, effective and responsible manner. It involves technical, cognitive and social abilities.
2. In the Information Age vast amounts of data are produced and shared instantly. Owning a device does not mean you can judge or use this information well. Users must develop skills to understand, manage and use information wisely, and to tell reliable information from unreliable.
3. Qualitative data is descriptive and non-numerical; it explains why and how (e.g. students' opinions about online classes; observation notes). Quantitative data is numerical, countable or measurable; it is used for comparing and making charts (e.g. marks in a test; hours spent online).
4. An interview is a planned conversation with one person or a small group, giving mostly qualitative, in-depth answers. A survey asks the same set of questions to many people (on paper or via Google/Microsoft Forms) and is efficient for large groups; online surveys also collect responses and make charts automatically.
5. A prototype is a simple early version of a product, model or solution. Users try it and give feedback; this feedback is data used to improve the product before time and effort are spent on the final version.
6. A simulation is used when real experiments are costly, risky or difficult. Example: a flight simulator to train pilots, or a traffic simulation to test the effect of closing a road.
7. Purpose: primary data is prepared for a specific study; secondary data is reused for a new purpose. Control: the researcher has high control over primary data (questions and process) and limited control over secondary data because it already exists.
8. Any four: ask permission (consent); avoid sensitive personal information; do not record names unless necessary (anonymity); keep responses private; use data only for educational purposes.
9. Quotes " " find an exact phrase, e.g. "study habits" students. filetype: finds a file format, e.g. "academic performance" filetype:pdf gives only PDFs. site: limits to a type of website, e.g. digital distraction site:.edu searches only educational sites.
10. A digital artefact is a final product created with digital technology that answers the research question with evidence and conclusions. Options: digital report (MS Word/Google Docs), presentation (PowerPoint/Canva/Slides), infographic (Canva/Adobe Express), dashboard (Power BI/Tableau).

Long questions — model answers

- Eight steps applied to water wastage.** (1) **Define the question:** "How much water is wasted at the college taps during break time, and why?" (2) **Data type:** both – quantitative (number of taps left running, minutes running) and qualitative (students' reasons). (3) **Methods:** observation to count running taps; a short survey to learn habits and reasons. (4) **Sample:** all 12 taps observed over 5 break times; 30 students from different classes surveyed. (5) **Tools:** an observation sheet (Tap no., Time, Left running Y/N, Minutes) and a Google Form. (6) **Collect** carefully, at the same time each day, recording accurately. (7) **Store** data in Google Sheets and keep a backup copy (download or USB). (8) **Ethics:** get the principal's permission; the survey is voluntary and anonymous with a consent question; data used only for the project.
- Survey on travel to college (model).** *Consent:* "This anonymous survey is for a Class XI project on travel to college. Do not write your name. Answers will be kept private and used only for this project. Do you agree to take part? Yes / No." *Questions:* Q1 How do you usually travel to college? Walk / Bus / Rickshaw / Motorbike / Car / Other – quantitative (categories). Q2 How many minutes does your journey usually take? ____ – quantitative. Q3 How many days a week are you late because of travel? 0 / 1 / 2 / 3 or more – quantitative. Q4 "My journey to college is safe." 1 (strongly disagree) to 5 (strongly agree) – quantitative (rating). Q5 Do you travel alone? Yes / No – quantitative (count). Q6 What is the biggest problem you face while travelling? – qualitative. Q7 Suggest one way the college could make travel easier. – qualitative. Questions are unbiased, ask one thing each, and options do not overlap. *Sample:* about 40 Class XI students, roughly 10 from each section, to be representative. *Ethics:* consent, no sensitive information (no home address), no names (anonymity), private storage, educational use only. Data is collected through Google Forms and exported to Google Sheets for analysis.
- Spreadsheet problem.** (a) $\text{=SUM(B2:B11)} \rightarrow 2+4+3+5+6+4+3+4+1+4 = 36$.
 $\text{=AVERAGE(B2:B11)} \rightarrow 36 \div 10 = 3.6$ hours. Sorted: 1, 2, 3, 3, 4, 4, 4, 4, 5, 6; $\text{=MEDIAN(B2:B11)} \rightarrow (4 + 4) \div 2 = 4$. $\text{=MODE(B2:B11)} \rightarrow 4$ (appears 4 times). $\text{=MAX(B2:B11)} \rightarrow 6$; $\text{=MIN(B2:B11)} \rightarrow 1$. (b) $\text{=COUNTIF(B2:B11, ">4")}$ $\rightarrow$ values 5 and 6 $\rightarrow 2$ students; percentage $\text{=COUNTIF(B2:B11, ">4")}/\text{COUNT(B2:B11)} \rightarrow 2 \div 10 = 20\%$. (c) 1–2 hrs: 2 students (2, 1); 3–4 hrs: 6 students (4, 3, 4, 3, 4, 4); 5+ hrs: 2 students (5, 6). Check: $2 + 6 + 2 = 10 \checkmark$. A **bar/column chart** is suitable for comparing the groups (or a pie chart to show 20%, 60%, 20%).
- Twelve steps.** (1) Advanced searches with operators such as "mobile phone use" "study habits" students, "academic performance" filetype:pdf, digital distraction site:.edu; evaluate sources (author, recent, reliable site, facts/bias). (2) Identify data types: quantitative (hours daily, % distracted) and qualitative (reasons, suggestions). (3) Select method: survey via Google Forms/paper, because the same questions go to many students. (4) Prepare the survey: hours daily (quantitative), main activity (quantitative categories), how the phone helps (qualitative), one responsible-use suggestion (qualitative). (5) Sample: 40 Class XI students from different sections. (6) Collect ethically: consent, no sensitive data, anonymity, privacy, educational use only. (7) Secondary data: an article saying excessive screen time lowers concentration, to compare with results. (8)

Organise in Excel/Google Sheets (Student, Hours, Study Use, Main Activity, Distracted?), sort, total, average. (9) Charts: bar chart of daily phone use (1–3, 3–5, 5+ hrs). (10) Analyse: majority use 3–5 hours; half use for education; entertainment users report more distraction. (11) Conclusion: phones have positive and negative effects; should be managed wisely, not banned. (12) Recommendations: turn off social media notifications, set screen-time limits, take physical breaks. **Artefact:** one of report, presentation, infographic or dashboard, structured as Title, Research Question, Introduction, Method, Primary Findings, Secondary Summary, Charts, Conclusion, Recommendations, References.

5. **AI tools for homework (model).** *Survey:* consent statement (anonymous, voluntary, educational use only). Q1 Have you used an AI tool (e.g. ChatGPT, Gemini) for homework? Yes / No – quantitative. Q2 How many times a week? 0 / 1–2 / 3–5 / 6 or more – quantitative (non-overlapping categories). Q3 What do you use it for most? Explaining topics / Checking answers / Writing assignments / Translation / Other – quantitative (categories). Q4 "AI tools help me understand my subjects better." 1–5 rating – quantitative. Q5 Do you check AI answers against your textbook? Always / Sometimes / Never – quantitative. Q6 Describe one problem you have faced with AI answers. – qualitative. Sample: about 40 Class XI students from all sections. *Secondary sources:* "generative AI" students homework filetype:pdf; "AI in education" site:.edu; "artificial intelligence" policy site:.gov.pk; evaluate each for author, date, reliability and bias. *Artefact:* a **digital report** (MS Word/Google Docs) because the principal needs the full study – method, findings, charts, conclusion, recommendations and references – in a structured document; a one-page infographic could be attached as a quick summary. Charts would be made in Google Sheets from the form responses, and every visual must match the real data.

Self-assessment: I can...

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| <ul style="list-style-type: none"> ☐ define digital literacy and explain why it matters in the Information Age. ☐ name and explain the five data-collection strategies and choose the right one for a scenario. ☐ compare primary and secondary data by source, examples, purpose and control. ☐ follow ethical rules: consent, anonymity, privacy and educational use. ☐ use advanced search operators and evaluate whether a source is reliable. | <ul style="list-style-type: none"> ☐ classify data as qualitative or quantitative and explain the difference in a table. ☐ write clear, unbiased survey questions and fix leading, double-barrelled or vague ones. ☐ list the eight steps for designing a data-collection approach. ☐ use spreadsheet formulas such as =SUM() , =AVERAGE() and =COUNTIF() and calculate results correctly. ☐ choose a suitable tool (spreadsheet, presentation, infographic, report, dashboard) to present findings. |
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- ☐ describe the 12 steps of the digital inquiry project.
- ☐ plan a digital artefact using the ten-part standard structure.